24PY901
Quantum Principles, Structures and Computing

Module 2 – Unit 1

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Assignment 5: Gaussian wavepacket and its time evolution

Numerical simulation of a free particle

References: Griffiths, Chapters 2. Time Independent Schrödinger equation.

Gaussian wave packet [1]

A free particle has the initial wave function

$$\psi(x,0) = A \exp\left(-ax^2\right),$$

where A and a are constants (a is real and positive).

(1) **Normalized wave packet:** In the lecture, we have normalized $\psi(x,0)$ and found that

$$A = \left(\frac{2a}{\pi}\right)^{1/4}.$$

Practice: Graph the wavefunction using Python's Numpy, Scipy, and Matplotlib packages and convince yourself that the wavefunction is indeed normalized. Hint: Use quad() function of scipy.integrate module.

[1] (G) Problem 2.22

(2) **Coefficient spectrum:** The Gaussian wave packet can be considered to be linear combination of eigenfunctions $\exp(+\imath kx)$ with coefficient $\phi(k)$

$$\psi(x,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) \exp(+\imath kx) \, \mathrm{d}k,$$

where the coefficient $\phi(k)$ is inturn given by

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x,0) \exp(-\imath kx) \, \mathrm{d}x.$$

$\phi(k)$ is called the coefficient spectrum. In the lecture, we have found that

$$\phi(k) = \left(\frac{1}{2\pi a}\right)^{1/4} \exp\left(-\frac{k^2}{4a}\right).$$

Practice: Find the coefficient spectrum function numerically and validate it with analytical solution.

(3) **Time evolution:** Each eigenfunction evolves in time t as

$$\exp(+\imath kx) \quad \longrightarrow \quad \exp\left(-\frac{\imath E(k)t}{\hbar}\right) \exp(+\imath kx),$$

where $E(k)$ is the dispersion relation given by

$$E(k) = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}.$$

Since the coefficient spectrum is invariant, the wave packet at time t is given by

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) \exp\left(-\imath \frac{\hbar k^2 t}{2m} + \imath kx\right)$$

In the lecture, we have found that

$$\psi(x,t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{\exp\left(-\frac{ax^2}{1+\imath\frac{2\hbar a t}{m}}\right)}{\sqrt{1+\imath\frac{2\hbar a t}{m}}}.$$

Practice: Find the evolved wave packet numerically and validate it with the analytical solution.

(4) **Probability density:** The probability density function is given by

$$|\psi(x,t)|^2 = \left(\frac{2}{\pi}\right)^{1/2} w(t) \exp\left(-2w^2(t)x^2\right),$$

where

$$w(t) = \sqrt{\frac{a}{1+\left(\frac{2\hbar a t}{m}\right)^2}}.$$

Practice: Sketch $|\psi^2|$ (as a function of x) at $t = 0$, and again for some very large t. Qualitatively, what happens to $|\psi^2|$, as time goes on? Hint: The time scale of the system is

$$t_0 = \frac{m}{2\hbar a}.$$

(5) **Expectation value:** Find numerically $\langle x \rangle$, $\langle p \rangle$, $\langle x^2 \rangle$, $\langle p^2 \rangle$, Δ_x, and Δ_p and validate with corresponding analytical expressions

$$\begin{aligned}
\langle x \rangle &= 0, \\
\langle p \rangle &= 0, \\
\Delta_x &= \left(\frac{1 + \left(\frac{2\hbar a t}{m}\right)^2}{4a} \right)^{1/2}, \\
\Delta_p &= \hbar \sqrt{a}.
\end{aligned}$$

(6) **Uncertainty principle:** Does the uncertainty principle hold? At what time does the system come closest to the uncertainty limit?